

Motion Smoothing for Assistive Leader-Follower Robotic Arms

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Abstract

Teleoperation systems directly map human motion to remote manipulators, making performance highly sensitive to disturbances such as physiological tremor. While tremor suppression has been widely studied in high-precision domains such as surgical robotics, its behaviour within low-cost, leader–follower teleoperation systems remains under-explored.

This work presents a dual-arm teleoperation framework for evaluating tremor suppression strategies. A ROS2-based system using low-cost Interbotix PincherX manipulators was developed, incorporating synthetic tremor injection and modular filtering via low-pass and Kalman-based approaches. Performance was assessed through quantitative analysis of tremor-band attenuation and command latency, alongside a user study of perceived stability and responsiveness.

Low-pass filtering achieved the highest attenuation (>40 dB) but introduced substantial delay. In contrast, the Kalman filter provided lower attenuation (~ 19 dB) while maintaining significantly reduced latency. Despite its lower level of tremor attenuation, the Kalman filter was consistently perceived as more stable and responsive during user testing.

These results demonstrate that maximising attenuation alone does not improve teleoperation performance. Instead, responsiveness governs perceived stability, highlighting the need to balance noise suppression with real-time control fidelity. This work contributes both an accessible experimental platform and insight into the practical design of tremor suppression in teleoperation systems.

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Research Ethics Approval

This project received approval from the Informatics Research Ethics Committee.

Ethics Application Number: 736504

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The participants' information sheet and a consent form are included in the appendix.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Oliver Hill)

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Chapter 1

Introduction

1.1 Motivation

Teleoperation enables human interaction with robotic manipulators by mapping user motion directly to a remote system [1, 2]. This approach enables applications ranging from hazardous material handling to robot-assisted surgery [3, 4]. However, its effectiveness is fundamentally limited by the quality of human motor input. Physiological tremor and fatigue introduce involuntary oscillations into commanded motion, which propagate directly to the manipulator and reduce precision during fine control tasks [5, 6].

While high-end systems mitigate these effects through specialised hardware and advanced control architectures, such solutions remain largely confined to expensive and tightly controlled environments, including surgical theatres and industrial or research facilities [4, 7].

In contrast, emerging low-cost manipulator platforms offer a pathway towards more accessible and deployable robotic systems [8]. These systems are particularly relevant for domestic and assistive applications, where affordability and ease of deployment are essential for enabling real-world use outside specialised environments. However, tremor suppression in such resource-constrained settings remains under-explored, creating a gap between system accessibility and effective user control.

Suppressing tremor introduces a fundamental trade-off. Aggressive filtering attenuates high-frequency oscillations but introduces latency due to phase delay,

a time shift between input and output, degrading responsiveness and the operator’s sense of control [9]. This challenge is compounded in low-cost systems, where limited sensing and computational resources restrict the use of complex control strategies. As a result, existing approaches do not adequately balance suppression and responsiveness in accessible teleoperation platforms.

This thesis investigates how ergonomic controller design and modular tremor suppression algorithms can be integrated within a dual-arm teleoperated robotic system. The aim is to reduce involuntary input disturbance while preserving responsiveness during manipulation tasks. Rather than proposing a medical solution, this work develops an assistive teleoperation platform for systematically evaluating tremor suppression in practical settings.

1.2 Problem Formulation

The challenge addressed in this work can be framed as a control problem in teleoperated manipulation, where instability at the input is directly propagated to the remote system. For tasks requiring fine spatial precision, even small-amplitude high-frequency variations degrade control fidelity [10]. The key challenge is therefore to suppress these disturbances while preserving intentional motion.

A range of approaches to tremor mitigation exist, including mechanical methods and signal-processing techniques [10, 11, 12]. Within signal-processing strategies, methods may broadly be categorised into two distinct philosophies: direct frequency attenuation and model-based state estimation. To enable a focused and controlled comparison within an accessible teleoperation framework, this work selects one representative method from each category: low-pass filtering and Kalman-based state estimation (further expanded in Section 2.5).

These approaches embody fundamentally different strategies for addressing oscillatory disturbance: one prioritising simplicity and predictable attenuation, the other leveraging dynamic modelling to preserve intentional motion.

Furthermore, tremor suppression cannot be considered purely a software problem. The mechanical design and ergonomics of the controller influence how oscillatory motion is generated and transmitted. Evaluating filtering strategies therefore requires an integrated hardware–software framework capable of controlled exper-

imentation.

Accordingly, this thesis investigates the following research question:

How do low-pass filtering and Kalman-based state estimation compare in their ability to reduce oscillatory disturbance while preserving responsiveness within an assistive dual-arm teleoperated system?

To explore this question, the work examines:

1. How oscillatory input propagates through the teleoperation architecture.
2. The comparative attenuation achieved by low-pass and Kalman filtering under testing conditions.
3. The impact of each approach on task performance and perceived responsiveness.
4. The influence of controller ergonomics on prolonged usability.

1.3 Contributions

The primary contributions of this work are summarised as follows:

1. **Extension of an accessible dual-arm teleoperation platform.** An existing robotic platform based on modified Interbotix PincherX manipulators was extended to enable dual-arm teleoperation, ROS2 communication, and real-time tremor suppression using compact, low-cost hardware.
2. **Design and prototyping of a custom dual-arm teleoperation controller.** A physical input controller was designed and fabricated using iterative prototype 3D printing, enabling intuitive dual-arm control and human-in-the-loop evaluation of tremor effects.
3. **Development of a controllable tremor injection framework.** A joint-space disturbance model was implemented to simulate physiological tremor with bounded frequency, amplitude, and temporal variation. This enables controlled and reproducible evaluation of suppression strategies under consistent oscillatory conditions.
4. **Implementation and comparison of two tremor suppression approaches.** Two filtering strategies were implemented to reduce oscillatory

disturbances: a low-pass filter and a Kalman filter. This enables direct comparison of how simple signal smoothing and model-based estimation affect system stability and responsiveness under artificially injected oscillatory conditions.

5. **Development of an experimental framework for analysing input instability.** The combination of tremor injection, filtering methods, safety mechanisms, and recorded system data provides a structured framework for analysing how instability in human input propagates through teleoperation systems.

1.4 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 provides the necessary background, introducing key concepts in teleoperation, human motor tremor, and tremor suppression strategies. Relevant technologies utilised in the system, including robotic middleware, simulation tools, and hardware platforms, are also discussed to establish the technical context for the work.

Chapter 3 presents the implementation of the teleoperation framework. This includes the hardware platform and custom input controller, as well as the ROS2-based control architecture. Tremor injection and suppression modules, safety mechanisms, and simulation-based validation are then described.

Chapter 4 presents the evaluation of the system through quantitative and user-based testing. Experimental procedures for tremor injection and suppression are described, followed by analysis of attenuation performance and responsiveness. The chapter concludes with a discussion of the results, system limitations, and directions for future work.

Chapter 2

Background

This chapter outlines the concepts underpinning the proposed system, including teleoperation, tremor modelling, and signal filtering. These elements define how disturbances arise, how they can be represented, and how they may be suppressed within a dual-arm teleoperation framework.

2.1 Teleoperation Robotics

Teleoperation systems map human motion to a remote manipulator, making system performance directly dependent on input fidelity [2].

Modern teleoperation research focuses on maintaining stability and responsiveness under communication delay and uncertainty [13, 14]. However, these approaches often assume that operator input is stable. In practice, this assumption does not hold, as human motion contains structured disturbances that can degrade control performance.

Mapping Strategies in Leader-Follower Systems

Mapping between human input and robotic motion can be performed in joint space or task space [2, 15]. Joint-space mapping directly transmits joint angles, while task-space approaches compute an end-effector pose using inverse kinematics, which determines the joint configuration required to achieve a desired end-effector position and orientation.

Task-space methods are widely used where Cartesian accuracy is required. However, they introduce additional computation and are sensitive to modelling errors, singularities, and solver-induced latency [16, 17].

Joint-space mapping preserves a direct correspondence between input and actuator motion, avoiding these issues. This makes it more suitable for analysing how disturbances propagate through the system. Accordingly, this work adopts a joint-space mapping strategy.

Regardless of the mapping approach, disturbances in the input signal are transmitted to the manipulator. Understanding their characteristics is therefore essential.

2.2 Spectral Characteristics of Human Tremor

Physiological tremor is an oscillatory motion present during voluntary movement, typically occupying a frequency range of approximately 6–12 Hz in the hand and arm [18]. In teleoperation, this behaviour can be modelled as a disturbance superimposed on intended motion:

$$\boldsymbol{\theta}(t) = \boldsymbol{\theta}_{\text{voluntary}}(t) + \boldsymbol{\theta}_{\text{disturbance}}(t)$$

where $\boldsymbol{\theta}(t) \in \mathbb{R}^n$ represents the vector of joint positions in radians.

This decomposition reflects a common signal-processing interpretation, where observed motion is treated as the combination of a desired component and an additive disturbance. In this work, analysis focuses on a narrower 6–10 Hz band. This restriction is imposed by actuator and sampling limitations at higher frequencies, but still captures the dominant tremor components relevant to manipulation tasks.

Unlike random noise, tremor exhibits structured, band-limited behaviour [6, 12]. This predictable spectral structure makes it amenable to modelling and targeted suppression within robotic control systems.

2.3 Modelling Tremor in Robotic Systems

While more detailed models represent tremor as a stochastic, multi-frequency process, a simplified narrow-band approximation is often adopted for real-time applications [19, 20]. In this work, tremor is modelled as a sinusoidal disturbance superimposed on voluntary motion:

$$\theta(t) = \theta_{\text{voluntary}}(t) + A \sin(2\pi ft)$$

where A denotes the tremor amplitude and f the dominant oscillation frequency.

This formulation represents a deliberate first-order approximation, capturing the dominant frequency component of tremor while neglecting higher-order harmonics and stochastic variation present in physiological signals.

Artificial tremor injection is widely used to evaluate suppression strategies [21, 6], allowing frequency and amplitude to be varied systematically. This enables controlled and reproducible comparison of filtering approaches, isolating the effect of specific disturbance characteristics on teleoperated performance.

With an artificial model for tremor injection established, the following section considers approaches for mitigating its effect within teleoperation systems.

2.4 Tremor Suppression Approaches

Tremor mitigation can be applied either at the mechanical interface or within the control signal. The following outline these approaches and assess their applicability within the context of teleoperation systems.

Mechanical and Actuation-Based Mitigation

Mechanical approaches introduce damping or compensatory actuation at the limb, reducing oscillation before transmission [10, 22]. While effective, these methods are often task-specific and can introduce additional inertia or resistance.

In teleoperation, motion is transmitted digitally, meaning disturbances propagate through the command signal itself. Mechanical mitigation at the input device therefore does not directly address how disturbances are represented within the control pipeline.

Digital Tremor Suppression Strategies

In contrast to mechanical mitigation, digital approaches operate directly on the transmitted signal.

Frequency-based filtering exploits the separation between voluntary motion and tremor, attenuating high-frequency components while preserving lower-frequency behaviour. A first-order low-pass filter was used as a baseline due to its simplicity and predictable attenuation characteristics [6, 20]. However, such filters introduce phase lag, which can degrade responsiveness during teleoperation.

Model-based approaches, such as the Kalman filter, estimate the underlying voluntary motion using a dynamic system model and uncertainty weighting [23, 24]. In this work, the Kalman filter was selected due to its ability to provide real-time state estimation under noisy measurements while maintaining low computational overhead. Unlike purely frequency-based methods, it incorporates both system dynamics and measurement uncertainty, enabling more informed separation of voluntary motion and tremor. However, filter performance is dependent on the selection of model and noise parameters, requiring careful tuning to achieve an appropriate balance between tremor suppression and responsiveness.

Evaluating this distinction requires a consistent framework for quantifying suppression performance. The following section therefore defines the metrics used to measure tremor attenuation within the teleoperation system.

2.5 Quantifying Tremor Suppression Performance

To evaluate tremor suppression methods, it is necessary to quantify the reduction in oscillatory disturbance. In discrete-time signal processing, attenuation is commonly assessed using frequency-domain metrics [9], where attenuation refers to a filter's ability to reduce unwanted signal components.

Following standard spectral analysis methods [9], tremor attenuation is evaluated by analysing signal energy within the tremor band. Given a discrete-time signal $x[n]$, representing the sampled position of an individual joint from the state vector θ , its spectral content can be represented via the Fourier transform. The corresponding power spectral density (PSD) $P(f)$ describes the distribution of signal energy across frequency. The tremor energy is then defined as the band-limited

spectral energy:

$$E_{\text{tremor}} = \int_{f_1}^{f_2} P(f) df,$$

where $[f_1, f_2]$ corresponds to the tremor frequency band.

In practice, signals are discretely sampled, and the Fourier transform is approximated using the discrete Fourier transform (DFT). The band-limited energy is therefore computed as a sum over discrete frequency bins within the tremor band, providing a numerical approximation of the continuous formulation.

Attenuation is expressed in decibels (dB) as a logarithmic ratio of band-limited energy before and after suppression:

$$A_{\text{dB}} = 10 \log_{10} \left(\frac{E_{\text{filtered}}}{E_{\text{input}}} \right).$$

This metric provides a direct measure of tremor suppression by quantifying the reduction in oscillatory energy within the defined frequency band.

Having established how tremor suppression performance can be quantified, the following section introduces the middleware and system architecture used to implement and evaluate these methods within the teleoperation framework.

2.6 Robotic Middleware and System Architecture

Robotic middleware coordinates sensing, processing, and actuation within distributed systems. In this work, ROS2 enables the teleoperation pipeline to be decomposed into modular nodes for signal acquisition, tremor injection, filtering, and actuation, allowing suppression strategies to be inserted and evaluated under consistent conditions [25, 26].

The system was implemented using ROS2 Humble due to its compatibility with the Interbotix X-Series platform used in this work. The vendor-supported control stack and drivers ensure reliable communication with the robotic manipulators.

ROS2 also provides tools for data logging and validation, including `Rosbag` for recording timestamped message streams, which was used for experimental data collection. In addition, this project utilises ROS2 simulation tools prior to hardware deployment to enable safe testing.

2.7 Utilised Simulation Platforms

Simulation provides a controlled environment for evaluating robotic systems. In this work, the focus is on signal-level behaviour rather than physical interaction.

Gazebo provides high-fidelity physics simulation [27], but this is unnecessary for analysing joint-space signal propagation. Instead, RViz was used to visualise joint states directly, avoiding additional dynamics that could confound filtering behaviour [25]. This allows filtering performance to be evaluated in isolation, ensuring that observed effects arise from signal processing rather than simulated physics.

Following simulation-based evaluation, the system can be deployed on physical hardware.

2.8 Utilised Hardware Platforms

The system comprises of compact robotic manipulators, integrated actuation and sensing, and an embedded compute platform. Forming a cohesive architecture for dual-arm teleoperation and signal processing.

Follower Manipulators

The Interbotix PincherX-100 was selected as the follower manipulator. The physical platform was originally developed by Charvi Agarwal as part of a prior project and extended in this work to support dual-arm teleoperation and tremor suppression. The platform is typically described as a 4-degree-of-freedom (DoF) arm, referring to the revolute joints in the main kinematic chain [28]. In this work, gripper actuation is treated as an additional controllable DoF, giving a total of five controllable joints.

An additional `wrist_rotate` joint was physically introduced by Charvi Agarwal via daisy-chained actuation, in which the actuator is connected in series with the existing control and power chain, to increase end-effector orientation capability (as shown in Appendix Figure A.1). The modified platform therefore provides five pose-generating joints, with gripper actuation controlling grasp rather than spatial pose. This increases the range of achievable end-effector configurations while maintaining compatibility with the existing system architecture.

Joint Actuation and Sensing

Dynamixel actuators provide integrated control, position sensing, and communication over a shared serial bus [29], simplifying system integration. Alternative sensing approaches, such as potentiometers or standalone rotary encoders, were considered for leader joint measurement. While these options offer lower cost and reduced hardware complexity, they require additional circuitry for signal conditioning and lack native support for synchronised multi-joint communication.

In contrast, Dynamixel actuators provide high-resolution encoder feedback, built-in communication protocols, and daisy-chain wiring [29], enabling consistent and scalable acquisition of joint states across multiple DoF. This reduces integration effort and ensures reliable, synchronised data transmission within the teleoperation pipeline. This design choice prioritises system integration and data consistency over minimal hardware cost, aligning with the experimental objectives of the platform.

Embedded Compute Platform

A Raspberry Pi 4 was used as the central compute platform to execute ROS2 nodes and filtering algorithms. Compared to desktop systems, the Raspberry Pi provides a compact and low-cost solution suitable for embedded robotic applications, while still offering sufficient computational capability to support real-time message passing and filtering operations.

However, the use of an embedded platform introduces constraints on processing overhead and timing determinism, requiring careful management of node execution and communication within the teleoperation pipeline.

These hardware characteristics define the operational limits of the system and directly influence the design of the control pipeline.

2.9 Scope of the Experimental Platform

This work focuses on signal-level behaviour in assistive teleoperation within a home environment, examining how disturbances propagate through the control pipeline and how filtering affects stability and responsiveness. Although experiments are conducted using PX-100 manipulators, the methodology is not tied to

a specific platform. Instead, it provides a framework for analysing tremor suppression in joint-space teleoperation systems. Within this context, it is necessary to identify how existing approaches address tremor and where limitations remain.

2.10 Identified Research Gap

Teleoperation systems directly transmit human input, making performance dependent on input fidelity. Physiological tremor introduces structured, band-limited disturbances that degrade this fidelity, and existing suppression approaches broadly fall into frequency-based filtering and model-based estimation [6]. However, the interaction between tremor characteristics, these strategies, and assistive teleoperation system design remains insufficiently explored.

Prior work on tremor suppression is largely concentrated in medical and assistive robotics, often evaluated within specialised platforms. For example, Jia et al. [6] demonstrate effective tremor filtering in robotic-assisted surgery under tightly controlled conditions. In contrast, teleoperation research primarily focuses on communication delay, haptic feedback, and bilateral stability, often assuming stable operator input [13, 15, 30].

As a result, comparative evaluation of digital tremor suppression methods within accessible teleoperation systems remains limited. In particular, few studies assess frequency-based filtering and model-based estimation under shared conditions or provide reproducible joint-space disturbance frameworks for controlled evaluation.

This gap motivates the research question introduced in Chapter 1. To address it, this work develops a modular teleoperation platform supporting controlled tremor injection and structured comparison of low-pass filtering and Kalman-based estimation, enabling analysis of both signal stability and perceived responsiveness within a unified framework.

Chapter 3

System Implementation

This chapter presents the implementation of the teleoperation platform, combining a modified dual-arm hardware system with a ROS2-based software architecture for deterministic command transmission, tremor injection, and filtering.

The chapter begins with the hardware platform and actuator configuration, followed by the ROS2 system architecture and teleoperation bridge. Tremor injection and suppression modules are then introduced, alongside workspace safety mechanisms. Finally, simulation-based validation is presented.

3.1 Hardware Platform

The hardware platform was designed to support dual-arm teleoperation while maintaining portability and accessibility. Particular emphasis was placed on ergonomic input devices and reliable communication with the follower manipulators. As shown in Fig. 3.1, the system comprises of three primary subsystems: the leader controllers, a central compute platform, and the follower manipulators.

Joint measurements are sampled by leader-side microcontrollers and transmitted to a Raspberry Pi 4, which executes the teleoperation pipeline, including tremor injection, filtering, and safety enforcement. Processed commands are then forwarded to the follower manipulators via a Dynamixel driver board. Communication is handled over a daisy-chained Transistor-Transistor Logic (TTL) bus, enabling multi-joint control with simplified wiring. Independent power supplies are used across subsystems to maintain voltage stability during dual-arm operation.

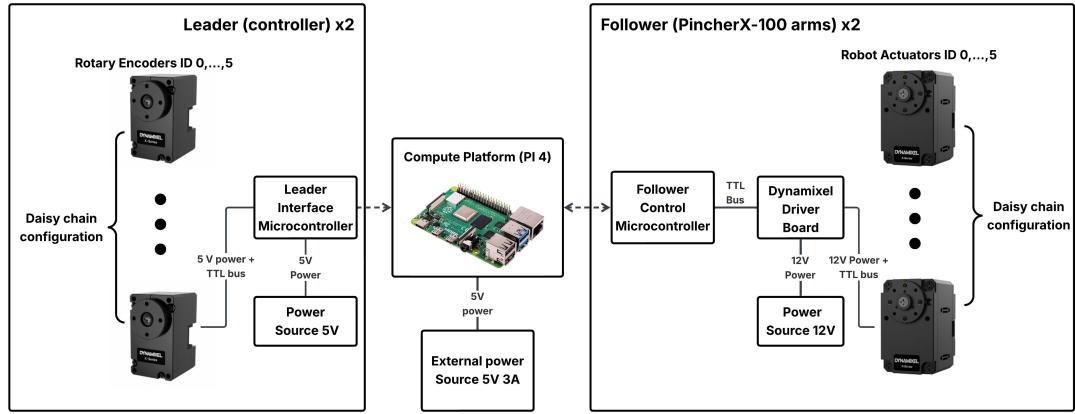


Figure 3.1: Hardware architecture of the dual-arm teleoperation platform. Dashed lines indicate directional serial communication.

The platform was constructed using low-cost, commercially available components. A detailed cost breakdown is provided in Appendix A.4, with the complete dual-arm system costing under £2000. This is substantially lower than high-end surgical robotic systems, such as the da Vinci platform, which can exceed \$1 million [31]. While such systems offer advanced sensing, precision, and clinical validation, the platform developed in this work demonstrates that key aspects of tremor-aware teleoperation can be explored using low-cost and accessible hardware. The physical realisation of this architecture is shown in Fig. 3.2.

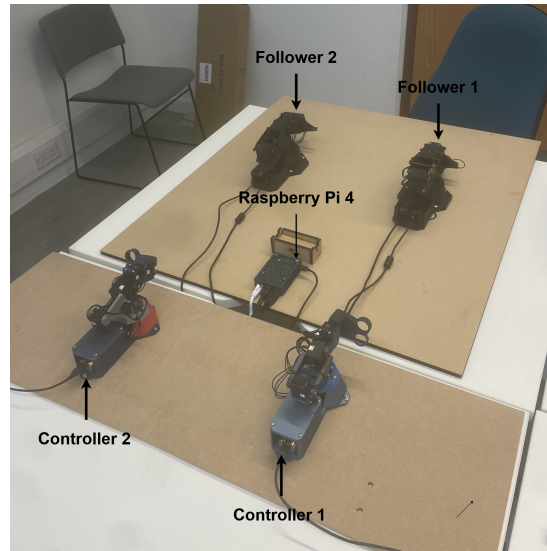


Figure 3.2: Experimental setup of the dual-arm teleoperation system.

Notably, the majority of system cost is attributed to the manipulators themselves,

with control hardware contributing a relatively small proportion of under £300. For comparison, low-cost teleoperation interfaces such as the GELLO system are reported at approximately \$300 (~£240) [8]. This indicates that the input hardware developed in this work is comparable in cost to existing low-cost interfaces, while being integrated into a complete dual-arm teleoperation system.

Manipulator Software Integration

The follower manipulator software stack was extended to support the additional `wrist_rotate` actuator without modifying the unified robotic description format (URDF), a standard text-based model used to define a robot’s kinematic structure. Instead, integration was achieved by updating the Dynamixel motor configuration files, incorporating the actuator into the existing control group and enabling it to be addressed within the ROS2 communication pipeline. This preserved compatibility with the Interbotix framework while enabling seamless inclusion of the additional joint in joint-space control.

Additional modifications were required to support dual-arm operation. Separate motor configuration files were defined for each manipulator to accommodate distinct motor ID assignments, allowing both arms to be controlled independently within the same system.

Actuation parameters were tuned to balance responsiveness and controllability. High velocity and acceleration values led to unstable motion, while low values introduced noticeable lag. The final configuration provided smooth and responsive behaviour without compromising stability across both manipulators, forming a reliable basis for subsequent leader-side control. The corresponding parameter configuration is provided in Appendix A.5.

Leader Controller Design

Two custom leader manipulators were developed to capture operator motion and provide joint-space input. Each controller implements six DoF, enabling direct mapping to the follower manipulators.

Position Sensors

A primary design requirement for the leader device is low input impedance, ensuring joints can be easily back-driven by the user. Here, back-drivability refers to the ease with which an actuator can be manually rotated against its internal gearing. Actuator selection therefore required balancing angular resolution, defined as the smallest measurable change in joint position, against the resistance introduced by internal gearing.

Candidate Dynamixel actuators were evaluated, as summarised in Table 3.1. Higher gear ratio models, such as the XL330-M288-T and XL430 series, offer increased resolution but introduce significant resistance to manual motion, reducing usability for teleoperation.

Table 3.1: Comparison of Candidate Dynamixel Actuators for Leader Joint Design

Model	Gear	Effort	Res.	Size	Suitability
XL330-M077-T	77:1	Easy	Mod.	Compact	Selected
XL330-M288-T	288:1	Difficult	High	Compact	Moderate
XL430-W250-T	~250:1	Very Diff.	High	Larger	Poor
XM430-W210-T	~210:1	Very Diff.	High	Large	Poor

The XL330-M077-T actuator was selected as it provides sufficient resolution while remaining easily back-drivable, allowing smooth and responsive user input.

To improve ergonomics, the controller structure was reduced in scale relative to the follower manipulators. Joint spacing was shortened by approximately 30%, producing a compact device that can be comfortably operated with one hand.

Additive Manufacturing

Custom links and mounting brackets were designed in TinkerCAD and fabricated using 3D printing. Its cloud-based workflow supports rapid iteration and is well suited to the simple link geometries of the controller, which do not require advanced parametric modelling, where geometry is defined by adjustable parameters and constraints.

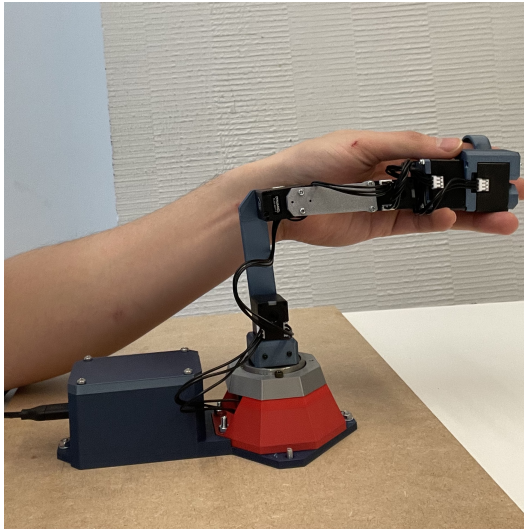
Material selection prioritised sufficient stiffness, ease of prototype printing and minimal deformation. Candidate materials were evaluated (Table 3.2), with PLA

selected due to its high printability and low warping. Although ABS and nylon offer greater strength, their increased print complexity was unnecessary for the low-load operating conditions of the leader device.

Table 3.2: Comparison of Candidate Materials for Leader Controller Fabrication

Material	Stiffness	Durability	Printability	Warping	Suitability
PLA	Moderate	Low	High	Low	Selected
PETG	Moderate	Moderate	Moderate	Low	Moderate
ABS	High	High	Low	High	Poor
Nylon	High	Very High	Low	High	Poor

Several design iterations refined the end-effector interface and improved structural rigidity. Initial vertical grip designs (Appendix Fig. A.3) were difficult to stabilise during operation. The final design introduces a widened support surface beneath the gripper, allowing the user to brace the controller while actuating the pinch mechanism. The resulting configuration is shown in Fig. 3.3.



(a) Operator holding leader controller in home position during grasping task.



(b) Operator opening leader controller end effector grips during grasping task.

Figure 3.3: Final configuration of the custom leader controller used for teleoperation input.

With the physical setup established, the system can now be considered from a

signal-processing perspective, where captured joint states are transmitted, modified, and enforced through the teleoperation pipeline.

3.2 Software Architecture

The teleoperation framework is implemented in ROS2 using a modular, communication-aware architecture. The data flow is summarised as follows:

1. Position sensors are sampled by the microcontroller and relayed over a serial interface.
2. Leader joint states are received by teleoperation bridge nodes.
3. Optional artificial tremor is injected in joint space.
4. A selectable filter (none, low-pass, or Kalman) processes the signal.
5. Commands are sent to a centralised safety node.
6. Workspace constraints are evaluated via forward kinematics to ensure end-effector positions remain within the defined workspace (outlined in Section 3.7).
7. Validated commands are forwarded in parallel to the follower manipulators.

This structure enables controlled comparison of suppression strategies under consistent communication conditions, while maintaining architectural simplicity. Figure 3.4 illustrates the node-level architecture and data flow.

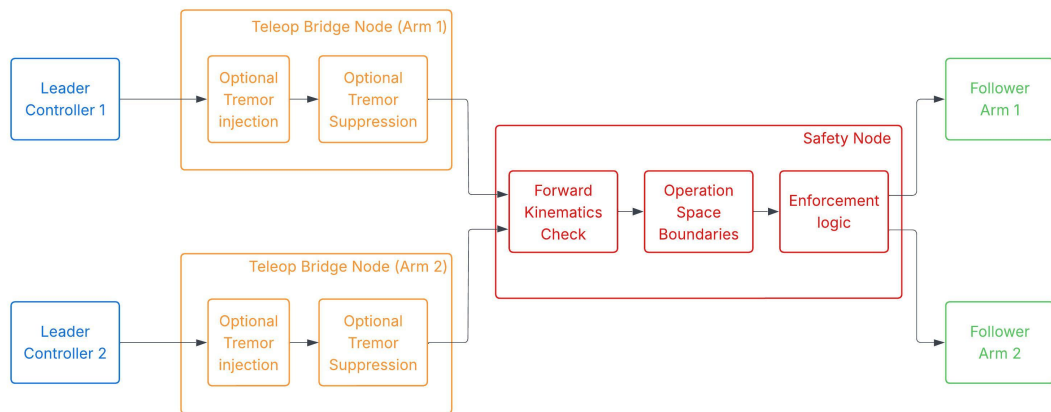


Figure 3.4: Software architecture of the dual-arm teleoperation system.

With the overall software architecture established, the following sections describe the implementation of its key components. These include leader joint acquisition, dual teleoperation bridges, tremor injection, tremor suppression modules, and the safety mechanisms responsible for enforcing workspace constraints.

3.3 Leader Joint Acquisition

Accurate fixed-interval sampling of leader joint states underpins the teleoperation pipeline. Each controller uses six XL-series Dynamixel actuators in position feedback mode. To minimise mechanical resistance and ensure low input impedance, actuator torque is disabled, allowing the motors to function as rotary encoders.

From a signal-processing perspective, the sampling frequency must satisfy the Nyquist criterion [9], which states that the sampling rate must be at least twice the highest frequency component present in the signal. The corresponding maximum representable frequency is

$$f_{Nyquist} = \frac{f_s}{2}.$$

If this condition is not met, higher-frequency components are misrepresented as lower-frequency oscillations, a phenomenon known as aliasing, distorting the tremor band and compromising evaluation. This imposes a trade-off: higher sampling rates improve frequency-domain accuracy but increase communication and processing overhead within the embedded ROS2 pipeline, introducing latency.

With $f_s = 22$ Hz, the Nyquist frequency is 11 Hz, slightly above the selected tremor band of 6–10 Hz (Section 2.2). This ensures tremor oscillations remain resolvable without aliasing while maintaining real-time performance within the constraints of the embedded platform.

Each actuator provides 12-bit encoder resolution (4096 counts per revolution) [32]. Raw readings p_i are converted to joint angles in radians as

$$\theta_i = \frac{p_i}{4096} \cdot 2\pi,$$

with values wrapped to $[0, 2\pi)$ to ensure numerical consistency. No additional transformation is applied, preserving a direct mapping between leader and follower joints.

The six joint angles are aggregated into the state vector

$$\theta = [\theta_1, \theta_2, \dots, \theta_6],$$

which is forwarded to the `teleop_bridge` node for command generation and optional disturbance processing.

3.4 Teleoperation Bridge Node

The central software components of the teleoperation pipeline are the `teleop_bridge` nodes. These nodes are responsible for translating motion captured from their respective leader controllers into joint commands for the follower manipulators, while also providing the insertion point for tremor modelling and suppression algorithms.

Joint Command Generation

Leader joint positions are acquired from the controllers through a serial interface and converted into relative joint displacements with respect to a calibrated home configuration. The follower manipulators initialise to a predefined home pose when powered for actuator safety, providing a fixed reference for commanded motion. In contrast, the leader controllers can be in arbitrary configurations at startup. As shown in Figure 3.5, during system initialisation the operator holds both controllers in a predefined pose while the system records encoder offsets. These offsets align the leader configuration with the follower home pose, allowing subsequent joint motion to be interpreted relative to this common reference.



Figure 3.5: Controller calibration procedure.

Once calibrated, the nodes continuously reads the encoder measurements, converts them into joint commands, and publishes them to the Interbotix control interface using the `JointGroupCommand` message type. Gripper commands are processed separately and transmitted using `JointSingleCommand` messages.

Follower Actuator Control Modes

The separation of arm and gripper commands reflects the different actuator control modes used by the system. The manipulator joints operate in *position control mode*, in which the actuator receives a desired joint angle and internally regulates the motor to reach that target position.

In contrast, the gripper is configured in *PWM control mode*, where commands correspond to a direct duty-cycle signal that drives the actuator without a fixed position target. This is advantageous for grasping tasks, as contact with an object can prevent the actuator from reaching a commanded position in position control mode, potentially leading to motor stall. PWM control instead allows the gripper to apply closing force without requiring a specific joint position.

Integration of Tremor Injection and Suppression

The tremor injection and suppression modules were integrated directly within the teleoperation bridge node, allowing signals to be injected, filtered, and forwarded within a single processing stage. This avoids additional publish–subscribe hops in the ROS2 pipeline, reducing communication overhead and preserving a consistent command rate with deterministic timing.

In contrast, many teleoperation systems apply tremor mitigation as an external filtering stage [13]. While this improves modularity, integrating processing within the control framework provides tighter control over signal timing and avoids additional middleware latency.

Within this architecture, tremor generation and suppression remain logically independent. The tremor module produces the disturbed signal, while suppression operates solely on the resulting command stream, ensuring performance reflects only the applied filtering strategy. The disturbance generation method is described in the following section.

3.5 Tremor Injection Module

Consistent with Section 2.2, tremor is modelled as a band-limited sinusoidal disturbance with time-varying frequency and amplitude. The frequency f is sampled within $[f_{\min}, f_{\max}]$, corresponding to the 6–10 Hz tremor band, while amplitude bounds $[A_{\min}, A_{\max}]$ are selected empirically to produce observable oscillations without dominating voluntary motion.

The disturbance is applied independently to each joint as:

$$\Delta\theta_i(t) = g_i A(t) \sin(\phi(t)),$$

where g_i is a joint-specific gain coefficient and $\phi(t)$ is the phase of the oscillation. Tremor characteristics may vary across joints, with amplitude differing between segments [33], motivating joint-dependent scaling.

The phase is updated recursively according to:

$$\phi(t + \Delta t) = \phi(t) + 2\pi f(t) \Delta t,$$

allowing efficient discrete-time evaluation.

To avoid discontinuities, amplitude and frequency are updated via linear interpolation over finite intervals. At randomly selected segment durations (1–3 s), new target values are sampled and blended into the active parameters, producing smooth quasi-stationary behaviour.

The disturbed command is then given by:

$$\theta_{\text{disturbed}} = \theta_{\text{clean}} + \Delta\theta,$$

where $\Delta\theta$ is the vector of joint perturbations.

This formulation provides controlled variation of amplitude and frequency, smooth temporal evolution, and deterministic reproducibility through seeded randomisation. The resulting signal is propagated through the teleoperation pipeline for optional suppression prior to execution.

3.6 Suppression Methods

The mathematical formulation of the filtering approaches implemented in this work follows the classical estimation framework described in [24].

Low-Pass Filter Implementation

To provide a frequency-based baseline for tremor suppression, a first-order low-pass filter was implemented within the teleoperation bridge node. Low-pass filtering attenuates high-frequency components while preserving lower-frequency motion [34].

A discrete-time exponential moving average (EMA) formulation was used:

$$y[k] = (1 - \alpha)y[k-1] + \alpha x[k]$$

where $x[k]$ is the input signal, $y[k]$ the filtered output, k the discrete time index, and $\alpha \in (0, 1)$ the smoothing coefficient.

$$\tau = \frac{1}{2\pi f_c}, \quad \alpha = \frac{\Delta t}{\tau + \Delta t}$$

The value of α governs the trade-off between attenuation and responsiveness. Smaller values of α produce stronger smoothing, increasing tremor attenuation but introducing greater delay, while larger values reduce delay at the cost of weaker suppression.

The cutoff frequency was selected based on the spectral separation between voluntary motion and tremor, with empirical testing indicating that 3 Hz provided the most effective balance between attenuation and responsiveness.

The filter operates independently per joint and introduces minimal computational overhead, making it well suited to real-time embedded execution. This method provides predictable frequency attenuation, serving as a baseline against which model-based approaches can be evaluated.

Kalman Filter Implementation

A discrete-time Kalman filter was implemented independently for each joint to provide a model-based alternative to frequency attenuation. The filter estimates underlying voluntary motion using a dynamic system model rather than explicitly restricting signal bandwidth.

Each joint is modelled using a constant-velocity state:

$$x_k = \begin{bmatrix} \theta_k \\ \dot{\theta}_k \end{bmatrix}, \quad x_k = Fx_{k-1} + w_k,$$

where θ_k and $\dot{\theta}_k$ represent joint position and velocity. The state transition matrix is

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}.$$

Process noise $w_k \sim \mathcal{N}(0, \mathbf{Q})$ accounts for unmodelled acceleration, enabling adaptation to changes in motion.

This formulation introduces a modelling assumption: voluntary motion is approximated as smooth and locally constant in velocity, while higher-frequency components are treated as disturbance. This simplification supports real-time estimation but may limit representation of rapid intentional motion. However, in assistive domestic teleoperation, user input is expected to be deliberate and relatively slow, making this approximation appropriate for the intended application.

The measurement model assumes direct observation of joint position:

$$z_k = Hx_k + v_k, \quad H = [1 \ 0],$$

with measurement noise $v_k \sim \mathcal{N}(0, R)$.

The filter recursively alternates between prediction and update steps, combining model-based estimates with incoming measurements.

Tuning Interpretation

Filter behaviour is governed primarily by the measurement variance R and the process acceleration variance σ_a^2 :

- R , controlling trust in the observed signal
- σ_a^2 , controlling responsiveness to motion changes

Higher R increases smoothing but introduces lag, while higher σ_a^2 improves responsiveness at the cost of reduced attenuation. These parameters therefore define the trade-off between tremor suppression and control responsiveness.

Parameters were tuned empirically across representative teleoperation trials to balance these competing objectives under both stationary and task-based conditions. Final values are provided in Appendix A.6.

With tremor injection and suppression integrated into the control pipeline, it is necessary to ensure that resulting commands remain within safe operating limits. The following section therefore introduces the safety constraints applied during physical experimentation.

3.7 Workspace Constraint Enforcement

Forward kinematics maps joint configurations to the corresponding end-effector position in Cartesian space. For each incoming leader controller joint command vector $\boldsymbol{\theta} \in \mathbb{R}^n$, the end-effector position is computed as

$$\mathbf{p}_{ee} = f(\boldsymbol{\theta}) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

In this context, forward kinematics is used to evaluate the spatial consequence of a given joint command. Inverse kinematics is not required, as the system does not generate joint configurations from desired end-effector poses, but instead validates existing joint-space commands.

Forward kinematics are evaluated using the Pinocchio library with the kinematics model derived from the PincherX-100 URDF. Only the joints contributing to the arm kinematic chain are included in this evaluation.

Task-Space Constraint Formulation

The permitted operating region is defined as the intersection of axis-aligned half-spaces in Cartesian task space.

A shared planar constraint is applied along the x -axis for both arms:

$$x \geq x_{min}.$$

This boundary prevents follower poses that are at risk of making contact with the user. Additionally, independent y -axis limits are applied to partition the workspace between the two manipulators:

$$\text{Arm 1 : } y \leq y_{max}^{(1)}, \quad \text{Arm 2 : } y \geq y_{min}^{(2)}.$$

These constraints prevent spatial overlap between the two follower arms and reduce the risk of cross-interference during dual-arm manipulation.

The resulting workspace for each arm is therefore defined as:

$$\Omega^{(i)} = \left\{ p_{ee} \in \mathbb{R}^3 \mid x \geq x_{min} \wedge y \in \Omega_y^{(i)} \right\}.$$

All bounds are exposed as ROS2 parameters, enabling adjustment without recompilation. Appendix Figure A.2 illustrates the resulting constrained operation regions.

Hard Constraint Enforcement

The barrier wall implements an impenetrable constraint via command gating rather than trajectory reshaping.

For each incoming command:

1. The end-effector position $\mathbf{p}_{ee}$ is computed.
2. If $\mathbf{p}_{ee} \in \Omega^{(i)}$, the command is forwarded and stored as the most recent safe command.
3. If $\mathbf{p}_{ee} \notin \Omega^{(i)}$, the system enters a blocked state and republishes the last safe command, effectively freezing motion at the boundary.

This approach preserves determinism and avoids introducing additional control dynamics into the teleoperation loop.

Hysteresis and Re-Entry Logic

Frequent switching between blocked and unblocked states was observed near workspace boundaries due to position sensor noise, resulting in unstable “chattering” behaviour. This is a common issue in hardware systems where small measurement fluctuations can repeatedly trigger constraint conditions.

To address this, a hysteresis margin $h > 0$ was introduced. While blocked, motion is only resumed once $x > x_{min} + h$ and the corresponding y-axis constraint is also satisfied with the same margin.

This ensures that the end-effector must re-enter the permitted region by a finite amount before control authority is restored, eliminating rapid switching and im-

proving boundary stability. The constraint behaviour was verified in simulation prior to deployment on the physical platform.

3.8 Simulation and Pre-Deployment Validation

Prior to physical deployment, the teleoperation framework was validated in simulation using RViz. This stage verified correct signal propagation and evaluated the workspace constraint enforcement before enabling hardware actuation.

Synthetic Teleoperation Interface

A synthetic teleoperation interface was developed as a ROS2 node to support controlled pre-deployment testing. The interface provides a graphical user interface (GUI) with direct slider-based control of each joint in radians, allowing independent operation of both manipulators during simulation (Fig. 3.6).

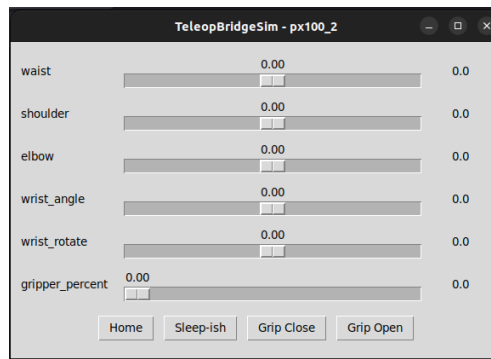


Figure 3.6: Tkinter GUI used during simulation.

This enables precise positioning, reproducible testing of boundary conditions, and rapid validation of joint configurations without requiring physical leader hardware. The node also retains tremor injection and suppression functionality, allowing baseline and filtered behaviour to be evaluated within a simulated control loop. By publishing to the same command topics as the teleoperation bridge, the interface acts as a deterministic input source for debugging and validation.

Barrier Wall Visualisation and Constraint Validation

Workspace constraints were validated in RViz through visualisation of the barrier geometry. A dedicated marker publisher was implemented using ROS2 visualisa-

This visualisation proved useful both for debugging and for validation. Because these markers were generated directly from the same parameters used by the safety node, the visualisation reflected the current constraint configuration exactly. It allowed end-effector motion to be inspected relative to the admissible workspace and confirmed that command freezing occurred at the expected geometric boundaries.

Simulation testing revealed a limitation in the initial forward-kinematics formulation used for workspace enforcement. The end-effector pose was computed using only the final actuator frame, neglecting the physical extension of the gripper. As a result, the safety constraint triggered too late, allowing the manipulator to visually penetrate the defined workspace in RViz.

To address this, the kinematic model was extended with an `ee_tool_length` parameter, applying a fixed offset along the end-effector x -axis to approximate the gripper reach. This modification aligned the evaluated pose with the effective operational point of the manipulator. As shown in figure 3.8, following this correction, barrier activation corresponded more closely to the physical extent of

the gripper, improving the validity of workspace enforcement.

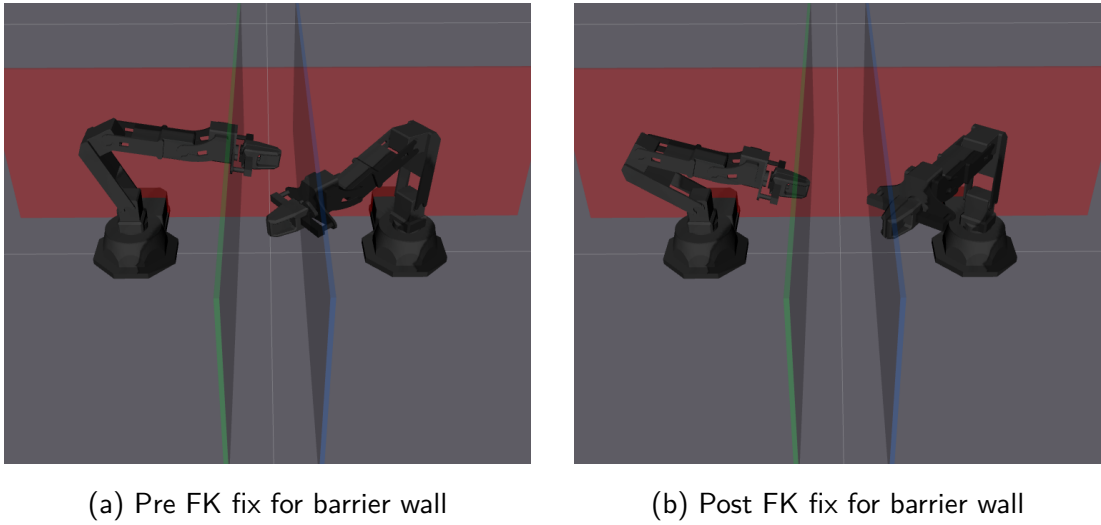


Figure 3.8: Comparison of workspace constraint behaviour before (left) and after (right) the forward kinematics correction.

With workspace constraints validated, the system was considered ready for experimental evaluation on the physical platform.

Chapter 4

Evaluations and Conclusions

This chapter evaluates the teleoperation framework and compares two tremor suppression strategies: low-pass filtering and Kalman-based state estimation. Evaluation is conducted through quantitative analysis of joint command data and a user-centred study, enabling assessment of both signal-level performance and its impact on operator interaction.

4.1 Evaluation Metrics

Tremor suppression performance was evaluated using two complementary metrics: attenuation of signal energy within the 6–10 Hz tremor band, and temporal delay, defined as the time shift between input and filtered output signals.

Attenuation (dB) quantifies the reduction in tremor bandpower between injected and filtered signals, computed using the formulation in Section 2.5.

Command latency (s) measures the delay between input and output signals, estimated via cross-correlation by identifying the time shift that maximises similarity between the signals.

Metrics were aggregated across joints within each trial, with reported values corresponding to the median of trial-level results, enabling direct comparison between filtering strategies.

4.2 Experimental Methodology

Experiments were conducted to evaluate system behaviour under multiple control conditions. Each trial followed a fixed sequence to assess both tremor suppression during static positioning and responsiveness during active motion.

Each trial consisted of three phases:

15s stationary → 30s task movement → 15s stationary

During stationary phases, the operator maintained a fixed pose to evaluate tremor suppression through analysis of oscillatory behaviour. The central task phase involved controlled manipulator movement to assess responsiveness and filter-induced latency, with consistent movement patterns used across trials to reduce variability between conditions.

Experiments were performed under three control conditions, all with synthetic tremor injection enabled:

- No suppression (tremor only)
- Low-pass filtering
- Kalman-based filtering

Joint command data were recorded using `rosvbag` and analysed to compute the defined metrics, enabling direct comparison of suppression performance and responsiveness across filtering strategies. Variability is represented using the interquartile range (IQR), shown as error bars in the figures.

Attenuation Performance

Figure 4.1 shows that both filtering approaches substantially reduce tremor-band energy compared to the unfiltered signal. The low-pass filter achieved the highest attenuation, with a median reduction exceeding 40 dB, while the Kalman filter provided a more moderate attenuation of approximately 19 dB.

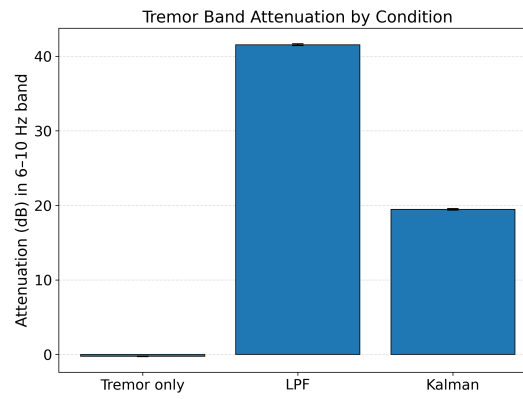


Figure 4.1: Tremor-band attenuation (6–10 Hz) across suppression methods.

Filter-Induced Delay

The improvement in attenuation achieved by the low-pass filter is accompanied by a substantial increase in response delay, as shown in Figure 4.2. The low-pass filter introduced a median lag of approximately 0.73 s with minimal variation, whereas the Kalman filter maintained a significantly lower delay of around 0.06 s, albeit with greater variability across trials.

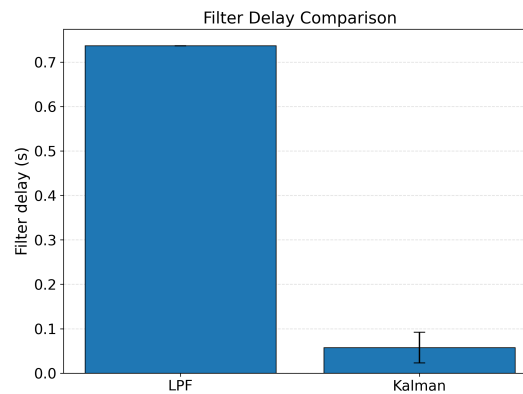


Figure 4.2: Measured command latency for each filtering strategy.

Signal-Level Behaviour

The qualitative differences between filtering strategies are shown in Figure 4.3. The low-pass filter produces a heavily smoothed signal with minimal high-frequency content, removing tremor at the cost of suppressing rapid motion changes. In contrast, the Kalman filter preserves signal structure while reducing oscillatory disturbance, enabling more responsive behaviour.

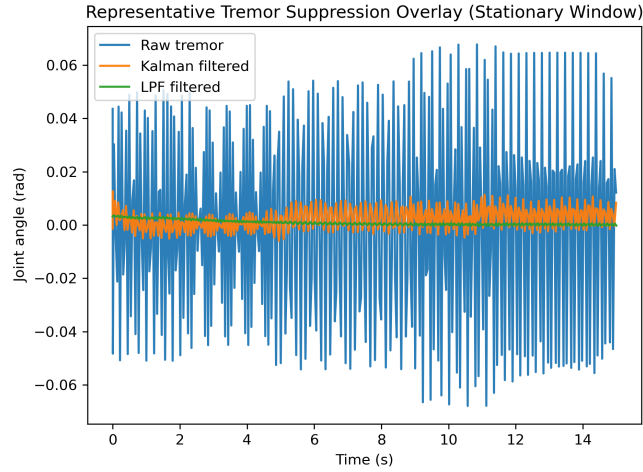


Figure 4.3: Representative joint signal showing tremor suppression behaviour.

Discussion

These results establish a clear trade-off between tremor attenuation and responsiveness. While low-pass filtering achieves higher suppression within the tremor band, it introduces significant delay that degrades real-time control. In contrast, the Kalman filter provides lower attenuation while maintaining near-immediate response.

This highlights a limitation of signal-level evaluation. Metrics such as attenuation quantify suppression effectiveness but do not capture the impact of latency on control fidelity, meaning methods that appear optimal in the frequency domain may perform poorly in interactive settings.

Accordingly, tremor suppression in teleoperation cannot be treated as a purely signal-processing problem. Responsiveness must be considered a primary constraint, as delays directly affect operator control.

To examine how these effects translate to usability, the following section presents a user-centred evaluation.

4.3 User Evaluation

A user study involving 14 participants was conducted to evaluate the impact of tremor suppression strategies on teleoperation usability. Participants performed a dual-arm manipulation task under three conditions: artificial tremor with no

suppression, low-pass filtering, and Kalman-based filtering. The task procedure and experiment setup is illustrated in Figure 4.4.

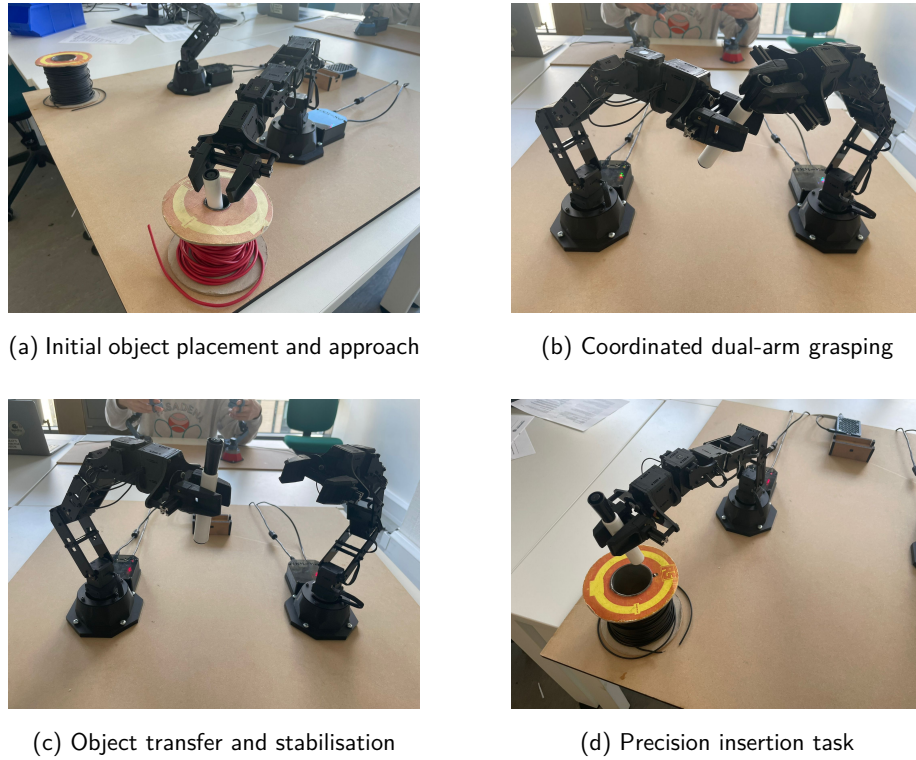


Figure 4.4: Sequential stages of the dual-arm teleoperation experiment.

These stages were designed to enforce bimanual teleoperation by construction, as the task could not be completed using a single arm alone. Users were therefore required to coordinate both manipulators simultaneously to achieve stable grasping, transport, and precise placement.

Each condition was evaluated using a 7-point Likert scale across perceived stability, responsiveness, and overall ease of control, to quantify qualitative opinions. Participants also ranked conditions based on stability and responsiveness. To minimise ordering effects, condition order was randomised between trials.

Observed behaviour

Figure 4.5 shows a divergence between perceived stability and responsiveness across conditions. The low-pass filter achieved the highest stability scores but was rated significantly lower in responsiveness, reflecting the impact of filter-induced delay. In contrast, the Kalman filter maintained high responsiveness

while achieving comparable perceived stability.

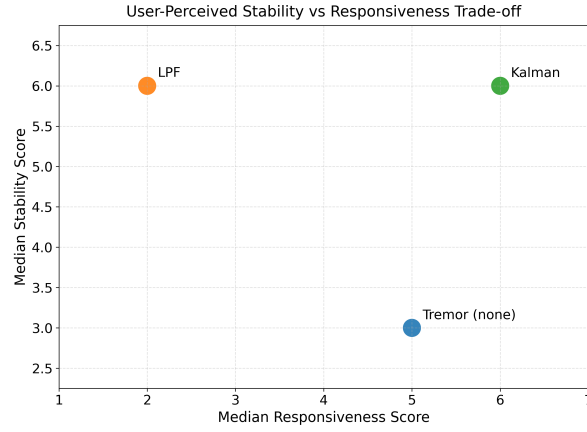


Figure 4.5: Median Likert scores for stability vs responsiveness.

Despite achieving approximately 20 dB greater attenuation, the low-pass filter was not consistently perceived as the most stable method. As shown in Figure 4.6, more participants ranked the Kalman filter as the most stable overall.

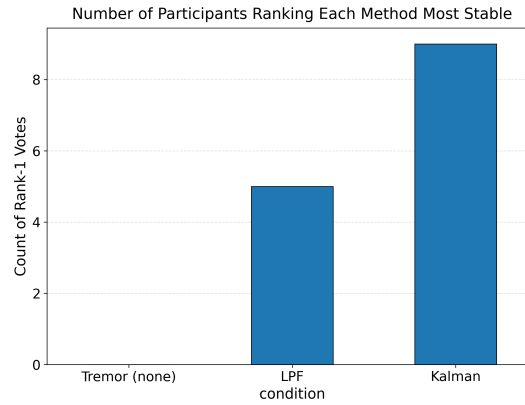


Figure 4.6: Number of participants ranking each method as most stable.

The unfiltered condition was consistently rated lowest in stability, particularly during precision tasks, confirming the disruptive effect of tremor on teleoperation performance.

Hardware and Workspace Constraints During Testing

During testing, actuator limitations were observed when manipulators were lifted from over-extended configurations, where insufficient torque resulted in temporary loss of controllability.

To ensure consistency across participants, the task was adapted to avoid these configurations by constraining the workspace, including reducing the separation between y-axis barrier walls to facilitate object transfer. This occasionally allowed limited overlap between the manipulators.

While these adjustments influenced the task setup, they did not affect the comparative evaluation of suppression strategies, as all conditions were tested under identical constraints.

Discussion

The user study reveals a mismatch between signal-level performance and perceived usability. Although the low-pass filter achieved substantially greater attenuation, it was not consistently perceived as more stable. This indicates that tremor suppression alone does not determine effective control.

Instead, perceived stability is governed primarily by responsiveness. While the low-pass filter effectively suppresses oscillatory disturbance, the associated delay disrupts the correspondence between user input and system response, reducing user confidence and producing a perceived loss of control. In contrast, the Kalman filter provides near-immediate feedback, which users interpret as increased stability despite lower attenuation.

These findings also highlight the role of the physical interaction interface. Qualitative feedback from participants (summarised in Appendix A.3) indicated that controller ergonomics influenced performance, with recurring issues including input coupling between joints, cable interference, and constrained hand positioning.

While the evaluation is based on subjective perception, the consistency of responses across participants indicates a clear preference for responsive behaviour over maximal attenuation.

4.4 Comparative Discussion

The results highlight a clear trade-off between tremor suppression and system responsiveness. Low-pass filtering achieves the highest attenuation at the cost of substantial delay, degrading real-time control. In contrast, the Kalman filter

maintains low latency while providing moderate suppression, resulting in improved user-perceived performance.

While the low-pass filter behaved in line with its expected frequency attenuation characteristics, the performance of the Kalman filter exceeded initial expectations in both attenuation and perceived usability. This can be attributed to the simplified sinusoidal tremor model adopted in this work, which aligns closely with the assumptions of linear state estimation. As a result, the Kalman filter is able to effectively separate voluntary motion from disturbance under these controlled conditions.

These findings demonstrate that perceived stability in teleoperation is governed primarily by responsiveness rather than absolute reduction in oscillatory disturbance. While increased attenuation improves performance initially, its benefit diminishes beyond a certain point. Once tremor is sufficiently suppressed to avoid disrupting task execution, further gains have limited impact on usability, and responsiveness becomes the dominant factor influencing perceived control.

Accordingly, tremor suppression in teleoperation should be treated as a multi-objective design problem. Metrics such as attenuation alone are insufficient to capture control fidelity, as they do not account for latency effects. Effective systems must balance disturbance reduction with real-time feedback, ensuring that improvements in signal stability do not come at the expense of operator control.

4.5 Limitations

While the developed system enabled controlled evaluation of tremor suppression strategies, several limitations must be considered when interpreting the results. These limitations arise from the physical interface design, modelling assumptions, and hardware constraints, highlighting areas for future refinement.

Human Interface Constraints

The custom-built leader controllers introduced limitations in ergonomic design and input decoupling. In particular, users reported that operating the scissor-style gripper mechanism could unintentionally induce rotation in the wrist joint,

indicating coupling between control inputs. This reduced precision during manipulation tasks and introduced additional variability in user performance.

Furthermore, the physical effort required to actuate the gripper was noted as a challenge by some participants, particularly those with reduced hand strength or dexterity. This suggests that the current controller design may not be suitable for a broad user population and limits the generalisability of the user study. These observations highlight the importance of ergonomic and accessible interface design in teleoperation systems, where input fidelity directly affects overall performance.

Tremor Modelling Assumptions

Tremor was modelled as a band-limited sinusoidal disturbance with time-varying amplitude and frequency. This enables controlled and repeatable evaluation, but represents a simplified approximation of physiological tremor. In practice, human tremor exhibits non-stationary and irregular characteristics [10, 33], with more realistic behaviour captured using biosignal recordings such as EMG or EKG [35].

This simplification is particularly relevant for model-based approaches such as the Kalman filter, whose performance depends on how closely model assumptions match the true disturbance. The sinusoidal approximation aligns with these assumptions, potentially contributing to the strong performance observed and overestimating effectiveness under real-world conditions.

Actuator and Hardware Limitations

The use of low-cost Dynamixel actuators introduced constraints on torque output and operational reliability. During user testing, motor shutdown occurred when participants attempted to recover the manipulators from over-extended configurations, resulting in temporary loss of controllability.

To maintain consistency across trials, the task was adapted to avoid such configurations. However, this constraint limits the operational workspace and highlights the sensitivity of the platform to load conditions. These hardware limitations also impact user confidence and system usability, particularly in tasks requiring sustained force or recovery from unstable poses.

Collectively, these constraints indicate that while the platform is suitable for

controlled experimental evaluation, further refinement of hardware, tremor modelling and interface design would be required for deployment in more demanding or real-world teleoperation scenarios.

4.6 Conclusion

This work investigated tremor suppression strategies within a dual-arm teleoperation system, focusing on the relationship between signal-level filtering performance and operator-perceived control.

Quantitative analysis showed that low-pass filtering achieved the highest attenuation of tremor-band energy (>40 dB) but introduced substantial delay, whereas the Kalman filter provided lower attenuation (~ 19 dB) while maintaining significantly lower latency.

User evaluation demonstrated that these signal-level improvements do not directly translate to improved teleoperation performance. Despite reduced attenuation, the Kalman filter was consistently perceived as more stable and responsive, indicating that perceived stability is governed primarily by responsiveness rather than absolute disturbance reduction.

These findings show that attenuation alone is insufficient for evaluating tremor suppression in interactive systems. Instead, effective teleoperation requires balancing disturbance reduction with real-time feedback, where latency directly impacts operator control.

The strong performance of the Kalman filter can be partly attributed to the simplified sinusoidal tremor model used in this work, which aligns closely with the assumptions of linear state estimation. As a result, the observed performance may overestimate effectiveness under real-world conditions.

Importantly, these results are achieved using low-cost hardware and lightweight computation, demonstrating that effective tremor-aware teleoperation can be realised without high-end sensing or intensive processing.

Overall, the results show that model-based filtering provides the most effective balance between suppression and responsiveness within the evaluated system, and establish responsiveness as a primary design constraint in human-in-the-loop teleoperation.

4.7 Future Work

Several directions for future work arise from the limitations identified in this study.

Firstly, the leader controllers should be improved to enhance ergonomics and input decoupling. Improving accessibility and reducing mechanical coupling between inputs would enable more reliable evaluation of teleoperation performance. Glove-based interfaces incorporating flex sensors or inertial measurement units (IMUs) offer a promising approach for more intuitive and low-resistance control [36].

Secondly, tremor modelling could be extended beyond the sinusoidal approximation used in this work. Incorporating physiological tremor data, such as EMG or EKG-derived signals [35], would enable evaluation under more realistic and clinically representative conditions.

Thirdly, adaptive tremor suppression strategies could be explored to balance attenuation and responsiveness in real time. Rather than using fixed parameters, adaptive filters adjust behaviour based on signal characteristics. For example, Araújo et al. [11] demonstrate adaptive methods for tremor suppression with improved tracking under varying conditions. Integrating similar approaches could enhance performance across both precision and dynamic tasks.

Fourthly, the hardware platform could be improved through the use of higher-torque actuators or more robust manipulators. This would expand the operational workspace, reduce failure under load, and improve overall system reliability during manipulation tasks.

Finally, collision avoidance can be extended beyond static workspace constraints by incorporating predictive detection between manipulators. By analysing joint trajectories, potential collisions can be identified in advance and commands adjusted accordingly, enabling safer and more flexible dual-arm operation.

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Appendix A

Supplementary Figures

A.1 Modified PincherX-100 Arm

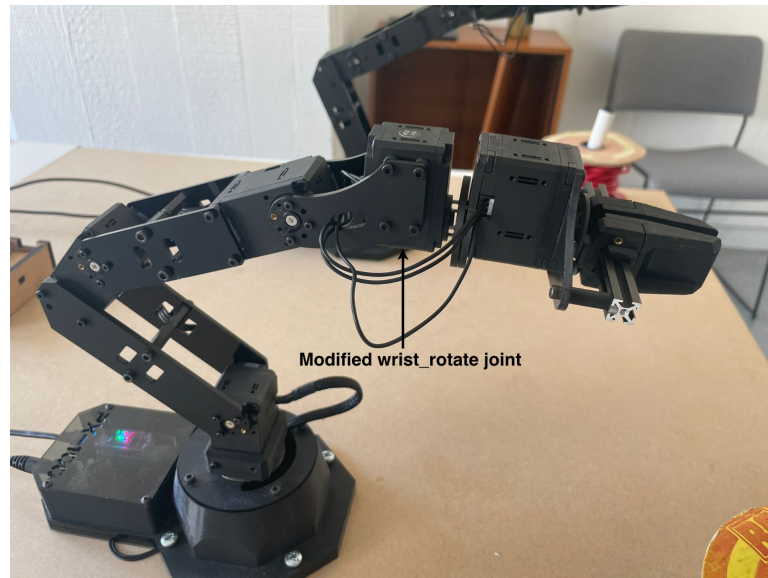


Figure A.1: Modified follower manipulator highlighting the wrist_rotate joint configuration. Hardware adjustments made by Charvi Agarwal were introduced to support the teleoperation setup and improve controllability of end-effector orientation during manipulation tasks.

A.2 Barrier wall CAD Visualisation

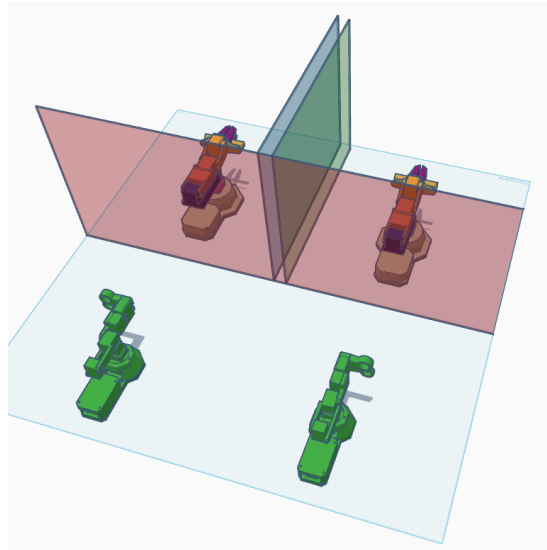
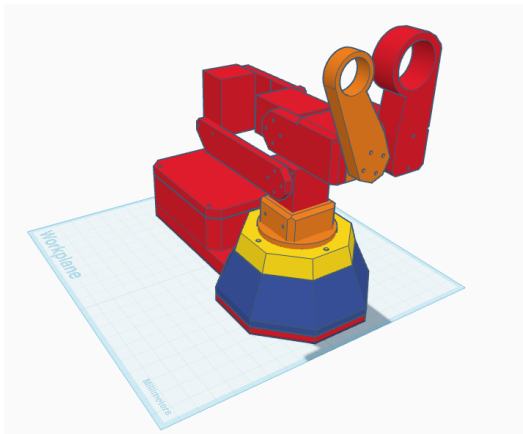
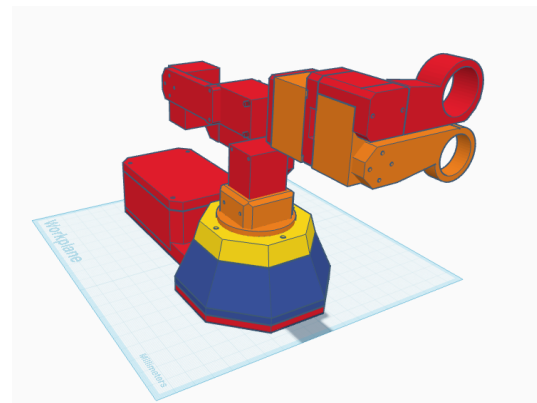


Figure A.2: Visual representation of the workspace constraints. The red plane denotes the shared x -axis barrier. The yellow and green planes represent the arm-specific y -axis limits for PincherX 1 (right) and PincherX 2 (left) respectively.

A.3 Design Iterations of System Controllers



(a) Design iteration 5: 5-DOF controller configuration



(b) Design iteration 9: 6-DOF controller configuration with wrist extension

Figure A.3: Early CAD design iterations of the leader controller. The initial design (left) utilised a 5-DOF configuration. A subsequent revision (right) introduced an additional wrist joint and more ergonomic orientation for hand-placement.

A.4 Dual-Arm System Price Breakdown

Table A.1: Hardware cost breakdown of the dual-arm teleoperation system. Prices sourced from RoboSavvy [37] and Interbotix documentation [28] including tax (March, 2026).

Component	Unit Cost (£)	Quantity	Total (£)
Interbotix PincherX-100 Arm	650.00	2	1300.00
Dynamixel XL430-W250 Servo	61.00	2	122.00
RB-150 Controller Board	29.10	2	58.20
Dynamixel XL330-M077-T Servo	18.50	12	222.00
Raspberry Pi 4 Model B	72.40	1	72.40
3D Printed Components (PLA, est.)	5.00	1	5.00
Total System Cost			1779.60

A.5 Actuation Configuration

The following pseudocode illustrates the actuator configuration used for velocity and acceleration tuning.

```

groups:
  arm:
    operating_mode: position
    profile_type: time
    profile_velocity: 800
    profile_acceleration: 200
    torque_enable: true

  singles:
    gripper:
      operating_mode: pwm
      torque_enable: true

```

A.6 Kalman Filter Parameter Configuration

The final parameter values used for the Kalman filter implementation are summarised in Table A.2. These values were selected through empirical tuning to balance tremor attenuation and responsiveness during teleoperation.

Table A.2: Kalman filter parameter configuration

Parameter	Symbol	Value
Measurement variance	R	2.5×10^{-3}
Acceleration variance	σ_a^2	2.0×10^{-1}
Initial position covariance	$P_{0,\theta}$	1.0
Initial velocity covariance	$P_{0,\dot{\theta}}$	10.0

A.7 Ergonomic Survey Results

This appendix summarises qualitative feedback on controller ergonomics collected from both structured user testing and informal evaluation during the dissertation fair. Comments were transcribed from participant responses and grouped into recurring themes to highlight common usability issues observed across users. The table presents representative examples illustrating each identified theme.

Table A.3: Summary of qualitative feedback on controller ergonomics (n=10).

Theme	Representative Comments
Input coupling	"Rotating wrist effects opening of pinchers and vice versa."
Hand positioning	"Turning inwards difficult from time to time, might dig in wrist."
Cable interference	"Cables got in the way."
Learning curve	"Learning curve to use, somewhat challenging in certain configurations to close pinchers."

Appendix B

Participation Information Sheet

Participant Information Sheet

Project title:	Motion smoothing for assistive leader-follower robotic arms
Principal investigator:	Garry Ellard
Researcher collecting data:	Oliver Hill

This study was certified according to the Informatics Research Ethics Process, reference number 736504. Please take time to read the following information carefully. You should keep this page for your records.

Who are the researchers?

1. Oliver Hill – Lead researcher (Beng Computer Science)
2. Garry Ellard – Supervisor (Student Technical Support Manager)

What is the purpose of the study?

This study aims:

1. To propose a platform for tremor suppression enabling the performance of high dexterity tasks.
2. Allow participants will operate a pair of robot arms through leader control devices.
3. Suppress tremor through tremor suppression methods, participants will be asked for their feedback on 3 tremor suppression algorithms.
4. Improve controller ergonomics through participant feedback.

Why have I been asked to take part?

You are an Informatics student, with an understanding of how to operate actuator technology reasonably and safely.

Do I have to take part?

No – participation in this study is entirely up to you. You can withdraw from the study at any time, up until submission of the survey without giving a reason. After this point, it will no longer be possible to withdraw because we are not collecting any data that would allow us to identify you.



What will happen if I decide to take part?

Testing interview process:

1. You will be invited to first explore using the pair of robot arms after calibration.
We will ask you for your feedback on ergonomics.
2. We will then ask you complete three high dexterity tasks through the robot arms in three different conditions:
 - a. No artificial tremor applied.
 - b. Artificial tremor applied.
 - c. Artificial tremor, and tremor suppression algorithm applied.
3. We will ask you to evaluate and compare 3 different tremor suppression algorithms.

Notes:

- The session will only occur once and not take longer than 30-45 minutes.
- Location will be the makerspace on the 3rd level of Appleton Tower.
- If operating the robot arms cause you discomfort in any way, please make our lead researcher aware, and we will end the interview early.

Are there any risks associated with taking part?

Robotic arms when used inappropriately can pose a significant danger to yourself and others in the area. It is essential you maintain a distance away from the arms during operation, as outlined by the researcher.

What data are you collecting about me?

The data we collect for our research is completely anonymous: We are not collecting any information that could, in our assessment, allow anyone to identify you. Your signed participant consent form will be kept separately from your responses and destroyed in May after research has concluded.

What will happen to the results of this study?

The results of this study may be summarised in published articles, reports and presentations. Your anonymised data may be published and can also be used for future research.



Who can I contact?

If you have any further questions about the study, please contact the lead researcher, Oliver Hill, s2157069@ed.ac.uk

If you wish to make a complaint about the study, please contact inf-ethics@inf.ed.ac.uk. When you contact us, please provide the study title and detail the nature of your complaint.

Alternative formats.

To request this document in an alternative format, such as large print or on coloured paper, please contact s2157069@ed.ac.uk



Appendix C

Participation Consent Form

Participant number: _____

Participant Consent Form

Project title:	Motion smoothing for assistive leader-follower robotic arms
Principal investigator (PI):	Garry Ellard
Researcher:	Oliver Hill
PI contact details:	Garry.Ellard@ed.ac.uk

By participating in the study, you agree that:

- I have read and understood the Participant Information Sheet for the above study, that I have had the opportunity to ask questions, and that any questions I had were answered to my satisfaction.
- My participation is voluntary, and that I can withdraw at any time without giving a reason. Withdrawing will not affect any of my rights.
- I consent to my anonymised data being used in academic publications and presentations.
- I understand that my anonymised data will be stored for the duration outlined in the Participant Information Sheet.
- I understand the risk associated with torque-actuated robot arms and will maintain an appropriate distance during operation, as outlined by the researcher.

Please tick yes or no for each of these statements.

1. I agree to being audio recorded.

☐	☐
Yes	No

2. I agree to being video recorded.

☐	☐
Yes	No

3. I allow my data to be used in future ethically approved research.

☐	☐
Yes	No

4. I agree to take part in this study.

☐	☐
Yes	No

Name of person giving consent

Date
dd/mm/yy

Signature



THE UNIVERSITY of EDINBURGH
informatics

Participant number: _____

Name of person taking consent

Date
dd/mm/yy

Signature



THE UNIVERSITY of EDINBURGH
informatics